

Tutorial

- 1) For an infinite square well, calculate the uncertainty product for the n th stationary state.
- 2) For a step potential, a) in case of $E < V_0$, determine the reflection coefficient and the decay distance inside the potential. b) In case of $E > V_0$, determine the reflection and transmission coefficient. c) Estimate the probability density in both the regions in both cases.
- 3) For a step potential, the wavefunction in region one outside the potential is superposition of incident and reflected wave and the region two inside the potential is an exponentially decaying wave. What is the probability current in both the regions? In case of a rectangular barrier, the region inside the barrier has two counter-propagating decaying waves. Determine the current density inside the barrier.
- 4) What is the minimum strength of a square well potential of height V_0 and width L so that it has at least 3 bound states?
- 5) For 3D rectangular potential box, with potential zero inside the box and infinite outside the box, use method of separation of variable to find solution to TISE. In case of cube, determine the energy and degeneracy of ground state and 1st excited state.